TOML-MIRI

Linear Algebra for Data Science — Exploration Exercises

October 2026

How to use these exercises

Each exercise is short (20–40 minutes). The aim is to explore, visualise and experiment until the concept makes sense; the code is only a means to that end. All exercises are in Python (NumPy, SciPy, Matplotlib; `ipywidgets` or Plotly for sliders), ideally in a Jupyter notebook. Each exercise has three parts:

- **Task** — what to compute and plot.
- **Explore** — what to change and watch.
- **Explain** — questions to answer in your own words. This is what you hand in.

Rules of the game:

1. Before running anything, write down what you expect to see. Then compare.
2. Never accept a plot you cannot explain. Go back to the lecture notes until you can.
3. Verify numerically: every identity you use (orthogonality, $A = U\Sigma V^\top$, $P^2 = P$, ...) gets a one-line `np.allclose` check.
4. Section numbers in brackets point to the lecture notes.

1. The four subspaces of a linear map [1.6–1.8.1]

1.1 See the four subspaces in 3-D

Task

Take a random 3×3 matrix of rank 2. Compute bases of its row space, null space, column space and left null space with the SVD. Plot each pair (row space + null space, column space + left null space) in an interactive 3-D figure.

Explore: Change the rank to 1 and to 3. Which subspaces grow, shrink or disappear?

Explain: Why is the null space always perpendicular to the row space? Check, for each case,

$$\dim \mathrm{RS}(A) + \dim \mathrm{NS}(A) = n, \qquad \dim \mathrm{CS}(A) + \dim \mathrm{LNS}(A) = m.$$

1.2 Where does A send a vector?

Task

For a rank-1 2×2 matrix, draw a grid of input vectors and their images Ax. Colour each input by its component along the null space.

Explore: Move a vector along the null-space direction. What happens to its image?

Explain: Why do all images land on one line? Relate it to the general solution [1.8.1]:

$$x = x_r + x_{ns}, \qquad Ax = Ax_r + \underbrace{Ax_{ns}}_{=0} = b.$$

1.3 Reproduce Example 1.12

Task

Compute the rank, the null-space vector and a column-space basis of

$$A = \begin{bmatrix} 2 & 1 & 1 & 1 \\ 3 & 0 & 0 & 2 \\ 1 & 3 & 6 & 0 \end{bmatrix},$$

using both `scipy.linalg.null_space` and the SVD.

Explain: Check every claim in the example numerically. Are columns a_2 and a_3 really linearly dependent? Which columns would you pick as a basis, and is the choice unique?

2. Positive definite matrices [2.6–2.7, 3.1, 3.5]

2.1 The shape of a quadratic form

Task

Plot the surface and contour lines of $f(x) = x^\top S x$ with $S = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$, with sliders for a, b, c. Show the eigenvalues and draw the eigenvectors on the contour plot.

Explore: Find slider settings giving a bowl (PD), a valley with a flat line (PSD), an upside-down bowl (ND) and a saddle (indefinite).

Explain: How do the signs of the eigenvalues predict the shape? Why do the contour ellipses align with the eigenvectors?

2.2 A sphere becomes an ellipsoid

Task

Apply a symmetric PD matrix $S = Q\Lambda Q^\top$ to points v on the unit circle step by step — first $Q^\top$, then Λ, then Q — and plot the three stages (Figure 5 in the notes).

Explore: Make one eigenvalue close to 0, then exactly 0.

Explain: What are the semi-axis lengths and directions of the final ellipse?

2.3 $A^\top A$ is always PSD

Task

Generate 1,000 random matrices A (some square, some tall, some rank-deficient) and record the smallest eigenvalue of $A^\top A$ for each. Plot a histogram.

$$x^\top A^\top A x = \|Ax\|_2^2 \geq 0.$$

Explain: When is the minimum exactly 0 (up to round-off)? Connect this to the proof in [2.7] and to the null space of A.

3. Multivariate Gaussian [3.6]

3.1 Build Σ from geometry

Task

Build a 2×2 covariance $\Sigma = Q\Lambda Q^\top$ from a rotation angle θ and two variances λ_1, λ_2 (sliders),

$$Q = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}, \qquad \Lambda = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}.$$

Sample N points from $\mathcal{N}(\mu, \Sigma)$, plot them with the ellipses $(x-\mu)^\top \Sigma^{-1}(x-\mu) = c^2$ for $c = 1, 2, 3$, and draw the eigenvectors scaled by $c\sqrt{\lambda_i}$.

Explore: $N = 100$, 1,000, 100,000 (as in the figure after 3.6.4). Set $\lambda_1 = \lambda_2$: what happens to the ellipses?

Explain: What fraction of points falls inside each ellipse? Compare with the χ^2 distribution with 2 degrees of freedom.

3.2 Generate correlated data from white noise

Task

Sample $z \sim \mathcal{N}(0, I)$, apply the transformation below, and compare with `np.random.multivariate_normal`.

$$x = \mu + Q\Lambda^{1/2} z \quad \Longrightarrow \quad \mathrm{Cov}(x) = Q\Lambda^{1/2} I \Lambda^{1/2} Q^\top = \Sigma.$$

Explain: Read the transformation as rotate–stretch–rotate [3.1]. Why is $\Lambda^{1/2}$ (and not Λ) the right stretch?

3.3 Recover the parameters

Task

Estimate μ and the sample covariance from the samples, then compare its eigenvectors and eigenvalues with the true ones as N grows.

$$\bar{x} = \frac{1}{N}\sum_{i=1}^{N} x_i, \qquad S = \frac{1}{N-1}\sum_{i=1}^{N} (x_i - \bar{x})(x_i - \bar{x})^\top.$$

Explain: Which is estimated worse, the direction of the long axis or the short axis? What happens when $\lambda_1 \approx \lambda_2$?

3.4 The precision matrix

Task

Using Σ from Example 3.3, plot the paraboloid (Figure 6) and the density surface side by side:

$$z = \tfrac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu), \qquad f_X(x) = \frac{1}{(2\pi)^{n/2}|\Sigma|^{1/2}}\, e^{-z}.$$

Explain: How are the eigenvalues of Σ and Σ^{-1} related? Why does the long axis of the ellipse correspond to the *small* eigenvalue of Σ^{-1}? Check $|\Sigma| = \lambda_1\lambda_2$ [3.6.6].

4. Singular value decomposition [4]

4.1 Rotate, stretch, rotate

Task

For a 2×2 matrix with entries on sliders, show the unit circle and its image in four panels: x, $V^\top x$, $\Sigma V^\top x$, $U\Sigma V^\top x$. Mark v_1, v_2 and their images $Av_i = \sigma_i u_i$.

Explore: Use $A = \begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}$ from Example 4.3. Make A singular. Make A orthogonal.

Explain: Read the spectral norm and the condition number directly from the picture [4.8, 4.9]:

$$\|A\|_2 = \sigma_{\max}, \qquad \kappa(A) = \frac{\sigma_{\max}}{\sigma_{\min}}.$$

4.2 SVD vs eigendecomposition

Task

Compute the SVD of the 3×2 matrix in Example 4.1, then the eigendecompositions of $A^\top A$ and $AA^\top$.

$$A^\top A = V\Sigma^\top \Sigma V^\top, \qquad AA^\top = U\Sigma\Sigma^\top U^\top, \qquad \sigma_i^2 = \lambda_i.$$

Explain: Match V, U and σ_i^2 with the eigen-results [4.2]. Why does $AA^\top$ have an extra zero eigenvalue? Are the singular vectors unique (look at their signs)?

4.3 Low-rank approximation of an image

Task

Load a grayscale image as a matrix, plot its singular values on a log scale, and show the rank-k approximations for $k = 1, 5, 20, 50, 100$ next to the original.

Explore: Compare a photo, a checkerboard, and an image of pure random noise.

Explain: Check the Eckart–Young errors [4.10]. Which image compresses best, and why?

$$A_k = \sum_{i=1}^{k} \sigma_i u_i v_i^\top, \qquad \|A - A_k\|_2 = \sigma_{k+1}, \qquad \|A - A_k\|_F = \Big(\sum_{i>k} \sigma_i^2\Big)^{1/2}.$$

4.4 Sum of rank-1 pieces

Task

Using the matrix in Example 4.6, plot each term $\sigma_k u_k v_k^\top$ as a heatmap, together with the running sum.

Explain: What does each rank-1 piece add? Compute the three Schatten norms of Example 4.4 and check them against the singular values:

$$\|A\|_2 = \sigma_1, \qquad \|A\|_F = \sqrt{\textstyle\sum_k \sigma_k^2} = \sqrt{\operatorname{tr}(A^\top A)}, \qquad \|A\|_N = \textstyle\sum_k \sigma_k.$$

5. Pseudoinverse [4.5]

5.1 Three roads to $A^\dagger$

Task

Compute $A^\dagger$ for a tall full-column-rank matrix, a wide full-row-rank matrix, and a rank-deficient square matrix, in the three ways below, and compare with `np.linalg.pinv`.

$$A^\dagger_{\text{left}} = (A^\top A)^{-1} A^\top, \qquad A^\dagger_{\text{right}} = A^\top (A A^\top)^{-1}, \qquad A^\dagger = V \Sigma^\dagger U^\top,$$

$$\Sigma^\dagger = \operatorname{diag}\left(1/\sigma_1, \ldots, 1/\sigma_r, 0, \ldots, 0\right) \in \mathbb{R}^{n \times m}.$$

Explain: Which formulas fail for which matrix, and why? Which one always works?

5.2 Left vs right inverse

Task

For the matrices in the previous exercise, compute $A^\dagger A$ and $A A^\dagger$.

Explain: Which product is the identity in each case? The other one is a projection matrix: check $P^2 = P = P^\top$ and say onto which subspace (row space or column space) it projects [2.3, 4.5].

5.3 Projection in pictures

Task

Take A from Example 2.2 (3×2). Plot its column space as a plane in 3-D, a vector t, its projection $\hat{t}$ and the residual e:

$$\hat{t} = Pt = A(A^\top A)^{-1} A^\top t, \qquad e = t - \hat{t}.$$

Explore: Move t with sliders.

Explain: Why is the residual always perpendicular to the plane? Which of the four subspaces does it live in?

6. Least squares for overdetermined systems [1.8.2]

6.1 Fit a line, see the geometry

Task

Generate 30 noisy points $y = 2 + 0.5t + \text{noise}$. Fit a line by solving the normal equations. Plot the data, the fitted line and the vertical residuals.

$$A = \begin{bmatrix} 1 & t_1 \\ \vdots & \vdots \\ 1 & t_{30} \end{bmatrix}, \qquad A^\top A \hat{x} = A^\top y, \qquad e = y - A\hat{x}.$$

Explain: Verify $A^\top e = 0$. In which subspace of A is e? Why is that the condition for the smallest $\|e\|_2$ [1.8.8]?

6.2 The error landscape

Task

Plot the cost below as a surface and contour map over (x_1, x_2) for the previous problem, and mark the least-squares solution.

$$J(x) = \|Ax - y\|_2^2, \qquad \nabla J = 2A^\top (Ax - y), \qquad \nabla^2 J = 2A^\top A.$$

Explain: Why is it a paraboloid? What do the eigenvalues of $A^\top A$ say about the shape of the contours? (Link to Section 2.)

6.3 Reproduce Example 1.14

Task

Solve the system below with the normal equations, `np.linalg.lstsq`, the pseudoinverse and the QR decomposition.

$$\begin{bmatrix} 2 & 1 \\ 1 & 3 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix}, \qquad \hat{x} = (A^\top A)^{-1} A^\top b.$$

Explain: Do all four agree with $\hat{x} = [0.9, 0.5]^\top$? Draw the three lines of the system in the plane and the least-squares point (Figure 4, right).

6.4 When the normal equations go wrong

Task

Fit a degree-10 polynomial with A built from columns $1, t, t^2, \ldots, t^{10}$, and compare the normal-equations solution with `np.linalg.lstsq`.

$$\kappa(A^\top A) = \kappa(A)^2.$$

Explore: Print `np.linalg.cond(A)` and `np.linalg.cond(A.T @ A)`.

Explain: Why does the identity above hold (use the SVD)? Why does it make the normal equations less accurate?

7. Minimum-norm solution of underdetermined systems [1.8.2]

7.1 One equation, two unknowns

Task

Plot the solution line of $2x_1 + x_2 = 3$, circles $\|x\|_2 = r$ with a slider for r, and the point

$$\min_x \|x\|_2^2 \ \text{ s.t. } \ Ax = b \quad \Longrightarrow \quad x_r = A^\top (AA^\top)^{-1} b.$$

Explore: Grow r until the circle first touches the line.

Explain: Why is the touching point the minimum-norm solution, and why is it perpendicular to the line (i.e. in the row space)?

7.2 Reproduce Example 1.13

Task

Compute x_r and the null-space vector, and plot in 3-D the solution line $x_r + c\,x_{ns}$.

$$\begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \end{bmatrix} x = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \qquad x_r = \tfrac{1}{30} \begin{bmatrix} 46 \\ -2 \\ -10 \end{bmatrix}, \qquad x_{ns} = \begin{bmatrix} 1 \\ -2 \\ 5 \end{bmatrix}.$$

Explain: Check $x_r^\top x_{ns} = 0$. Plot the function below: where is its minimum, and why?

$$g(c) = \|x_r + c\,x_{ns}\|_2^2 = \|x_r\|_2^2 + c^2 \|x_{ns}\|_2^2.$$

7.3 Gradient descent finds it too

Task

Run gradient descent from $x_0 = 0$ for a wide A, and plot the iterates in 3-D.

$$x_{k+1} = x_k - \eta\, A^\top (Ax_k - b).$$

Explain: Every update lies in the row space of A. Why does this make it converge to the minimum-norm solution? What happens if you start from a nonzero x_0?

8. Fourier basis and Fourier matrix [6]

8.1 Look at the Fourier matrix

Task

Build F_N for $N = 8$ and $N = 64$ from its definition. Plot the real and imaginary parts as heatmaps, and plot each of the first 8 columns as a sampled cosine/sine.

$$\omega_N = e^{j2\pi/N}, \qquad (F_N)_{ki} = \frac{1}{\sqrt{N}}\,\omega_N^{-ki}, \qquad k, i = 0, \ldots, N-1.$$

Explain: Verify $F_N^H F_N = I$. Why must you use the conjugate transpose and not the plain transpose? Which columns are the "low" and which the "high" frequencies?

8.2 DFT as a change of basis

Task

Compute $\hat{x} = F_N x$ for a random x and compare with `np.fft.fft(x)`.

$$\hat{x}_k = \sum_{i=0}^{N-1} x_i\, e^{-j2\pi ik/N}, \qquad x_i = \frac{1}{N}\sum_{k=0}^{N-1} \hat{x}_k\, e^{j2\pi ki/N}.$$

Explain: What scaling factor makes $F_N x$ and `np.fft.fft(x)` equal? Check the normalisations in [6.3.1] and the `norm="ortho"` option. With the unitary F_N, verify Parseval: $\|\hat{x}\|_2 = \|x\|_2$.

8.3 Speed of the FFT

Task

Time $F_N x$ (matrix–vector) against `np.fft.fft` for $N = 2^6, \ldots, 2^{14}$ and plot both on a log–log scale.

Explain: Do the slopes match the costs below [6.3.2]?

$$\text{matrix–vector: } O(N^2), \qquad \text{FFT: } O(N \log_2 N).$$

8.4 Circulant matrices are diagonal in the Fourier basis

Task

Build a random circulant matrix C (`scipy.linalg.circulant`), compute $F_N C F_N^H$, and show it as a heatmap.

$$C = F_N^H \Lambda F_N, \qquad Cx = h \circledast x = F_N^H \big(\Lambda\, F_N x\big).$$

Explain: Why is $F_N C F_N^H$ diagonal? Check that the diagonal equals the DFT of the first column h of C, and that Cx (a circular convolution) can be computed with two FFTs and one element-wise product.

8.5 Sparsity and denoising

Task

Reproduce Example 6.2: the signal below plus Gaussian noise. Plot the power spectrum, keep the k largest coefficients, invert.

$$f(t) = \sin(2\pi \cdot 50\, t) + \sin(2\pi \cdot 120\, t) + \varepsilon(t), \qquad \varepsilon(t) \sim \mathcal{N}(0, \sigma^2).$$

Explore: Vary k and the noise level σ. Repeat with pure white noise.

Explain: Why is the clean signal sparse in the Fourier basis but noise is not [6.4]? Compare the Fourier basis (universal) with the SVD basis of the image exercise in Section 4 (tailored to the data).